A Novel In Situ Dust Cover Index for Analyzing the Multispectral Camera Image Acquired by China's Zhurong Mars Rover

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***Abstract*— On May 15, 2021, China's first Mars rover, the Zhurong rover, successfully landed on the Utopia Planitia in the northern region of Mars. The multispectral camera (MSCam) onboard the rover has captured multispectral images, which provide spatial and spectral information about in situ observation targets and facilitate analysis of the types of materials on the Martian surface. However, frequent sandstorms on Mars are accompanied by dust deposition, and varying degrees of dust coverage have altered the original spectral characteristics of scientific detection targets, resulting in inaccurate material inversion. To address this issue, a novel in situ dust cover index (IDCI) is proposed. The data-driven method is based on the spectral features of dust cover in the MSCam multispectral bands. It provides a wealth of information through a simple, yet effective calculation that maximizes the discrimination between different categories of dust-impacted areas and estimates the degree of dust coverage. Experimental results obtained from 17 scientific observations by MSCam along the Zhurong rover's routing path confirm the effectiveness of the proposed IDCI. It effectively distinguished dust-free areas from invalid areas (e.g., shadows), while reflecting the degree of dust coverage in the scene. The IDCI demonstrated its superior performance, operating up to three times faster than other reference methods. Additionally, it exhibited a notable advantage over other techniques, achieving a variance ratio criterion (VRC) for target separation that was at least 5% higher. These results highlight the efficiency and effectiveness of the proposed IDCI, establishing it as a valuable tool for Martian surface analysis.**

***Index Terms*— Dust cover index (DCI), in situ detection, Mars, multispectral, multispectral camera (MSCam), Tianwen-1 (TW-1), Zhurong rover.**

I. Introduction

THE objective of Tianwen-1 (TW-1), China's first Mars exploration mission, is to utilize both an orbiter and a rover to investigate the potential existence of life on Mars where life might exist and the origin and evolution of Mars and to gain insights into the solar system [1]. On May 15, 2021, the Zhurong rover successfully landed in the southern part of the Utopia Planitia (109.925°E, 25.066°N), which is a large plain located in an impact basin with relatively smooth and flat terrain [2]. The flat, open topography of the region allows for the free flow of winds, which carry a considerable amount of dust particles, resulting in the formation of Martian dust cover. Furthermore, the seasonally strong winds and variations in low-pressure systems on Mars can contribute to the formation and propagation of dust storms [3]. The Martian surface is characterized by the presence of fine dust, which is a consequence of the planet's arid and frigid climate, its seasonal changes, and its erratic weather patterns [4]. The spatial and temporal scales of dust storms on Mars exhibit considerable variability, ranging from local events to global phenomena [5], [6], [7]. Dust-covered surface manifestations include extensive sand dunes and ripples, which are typical of Martian landscapes [8], [9], [10]. These features not only mask the underlying geological material but also complicate the interpretation of Martian surface properties.

Seasonal dust storms are a major natural phenomenon interfering with normal scientific exploration of Mars. The presence of airborne dust represents a significant challenge to Mars exploration, especially during the dust storm season when thick dust clouds engulf large areas of the planet [11]. In recent decades, Mars exploration missions have been conducted to study the morphology and mineralogical characteristics of the planet. These include the National Aeronautics and Space Administration (NASA) missions: the Mars Exploration Rover (MER) [12], the Mars Science Laboratory (MSL) [13], the Mars 2020 [14], the Mars Reconnaissance Orbiter (MRO) [15], and the Mars Global Surveyor (MGS) [16]. Additionally, the European Space Agency (ESA)'s Mars Express (MEX) [17] and China's TW-1 Mars Exploration Project [2] have also contributed to this field of study. In these missions, the payloads on orbiting spacecraft, landers, and rovers provide a wealth of scientific data for the study of various terrain features, material composition, and physical properties of planetary surfaces.

For orbiters remote sensing, the emissivity data provided by the MGS Thermal Emission Spectrometer (MGS-TES) was employed in the creation of a Martian global dust cover index (DCI) product. This product offers a more direct

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quantification of surface dust cover than is possible with measurements of thermal inertia or albedo [18]. The data product is of a large spatial scale but has a low resolution (3×6 km), which limits the ability to detect small-scale dust cover on Mars. In the TW-1 mission, the moderate-resolution imaging camera (MoRIC), the high-resolution imaging camera (HiRIC), and the Mars mineralogical spectrometer (MMS) were employed primarily for the purposes of conducting research into large-scale Martian topography, detecting mineral composition, and analyzing material distribution [19]. With regard to in situ and small-scale exploration of Mars by rovers, the Spirit, Opportunity, Curiosity, and Perseverance rovers have successfully conducted a series of scientific explorations on Mars, obtaining valuable multispectral data in the process. To study the types of Martian surface material and their distribution, the Zhurong rover was equipped with a scientific payload, the multispectral camera (MSCam), mounted on its mast. It provided high-resolution images of the landing and traverse areas in eight bands, ranging from the visible to near-infrared spectral wavelength [2].

However, the dust on Mars poses a major challenge that may hinder the achievement of scientific exploration objectives. The presence of fine dust particles on the Martian surface has the potential to obscure underlying minerals, thereby complicating the detection and analysis of key geological features. The presence of hydrated minerals can indicate the occurrence of past water activity. However, the identification of these minerals is more difficult when they are covered by dust. Dust deposition on the solar panels of the rover can significantly reduce its ability to generate electricity, which is a critical aspect of the rover's functionality. The Mars rover Zhurong encountered power supply issues due to the accumulation of dust on its solar panels, resulting in its prolonged inactivity in the Martian landscape. In addition, the presence of dust impedes the rover's movement and operational capabilities, causing slippage or loss of traction, and interfering with the functionality of mechanical components and instruments. Consequently, the planning and execution of exploration missions must consider the potential impact of dust cover on the Martian surface. To this end, the development of advanced techniques for studying the dust cover on the Martian surface is essential for enabling scientists to accurately select reliable scientific exploration targets.

Various approaches have been utilized to process Martian in situ multispectral camera data, including spectral parameterization, image transformation, and spectral matching and unmixing techniques. Spectral parameterization is a commonly used strategy due to its fast data-processing speed. This method entails the utilization of specific spectral parameters to derive meaningful information from the original multispectral bands. In this context, the red/blue band ratio is employed to estimate dust coverage and distinguish surface features of different targets [20], [21]. Other forms of spectral parameters are also calculated for analysis or validation to identify the rock and mineral types present on the Martian surface [22], [23], [24]. Image transformation methods, such as principal component analysis (PCA), minimum noise fraction (MNF), and decorrelation stretch (DCS), are crucial for reducing data dimensionality and emphasizing subtle spectral characteristics. Farrand et al. [25], [26] and Rice et al. [27], [28] used PCA to facilitate the identification of a diverse range of candidate spectral category pixels. MNF, which is similar to PCA, was utilized as a standardized program to identify the endmembers in multispectral images [29], [30]. The DCS technique, which is based on PCA from three selected image bands, is utilized to highlight subtle color contrasts in RGB composite images, thereby enabling more accurate discrimination between rocks and soil in the scenario [30], [31]. These methods are particularly useful for the creation of enhanced color images and decorrelation stretches, which provide a comprehensive assessment of spectral diversity in multispectral images.

Techniques based on spectral matching and unmixing, such as spectral mixture analysis (SMA), are employed for detailed spectral analysis. Farrand et al. [29] employed SMA to determine the identity of spectral endmembers in a multispectral scene and thus to assess the spectral variability of the scenario. This method is beneficial for the identification and quantification of the presence of diverse materials on the Martian surface. However, Bell et al. [32] has highlighted the potential for optically thin dust coatings on rocks to create areas that cannot be well modeled by end members, posing challenges for accurate analysis. Furthermore, dust deposition studies are conducted using calibrated target observations and a two-layer radiative transfer model to invert the results of dust deposition, including atmospheric optical thickness and deposition optical thickness. Additionally, the rate of dust deposition can be estimated, providing insights into the dynamic processes affecting the Martian surface [33], [34], [35].

In summary, spectral parameterization and image transformation methods are characterized by relatively fast processing speed and are commonly used in scenarios where landers or rovers must make quick operational decisions, such as driving, brushing, drilling, and so on. In contrast, spectral matching and spectral unmixing-based methods require more computational time and related initial settings.

Despite the success of the aforementioned techniques, several issues remain to be investigated further.

First, existing transformation-based methods (e.g., PCA, MNF, and DCS) are inadequate to explore the specific spectral features of Martian dust embedded within the multispectral data. These traditional methods typically emphasize data compression and noise reduction and tend to focus on broader geological features rather than the distinctive spectral characteristics of dust across different multispectral channels.

Second, the current methods for analyzing Martian dust cover, such as PCA and MNF, usually require prior knowledge and empirical judgment. These techniques must identify the principal components most closely associated with Martian dust coverage for further analysis. Moreover, the method based on DCS requires manual selection of input bands and subsequent visual interpretation of the stretching results, which is essential for determining dust coverage. This dependency on manual input limits the potential for these methods to be employed in a fully automatic fashion. Furthermore, these methods involve complex computations, including

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the manipulation of covariance matrices, eigenvalues, and eigenvectors, which are both computationally intensive and time-consuming when applied to large datasets. Therefore, they are not suitable for real-time, large-scale data processing.

Third, the current approaches are inadequate for providing a comprehensive account of the information on the Martian dust cover and its extent. Some existing research offers only DCS false-color composite images or principal component transformation results [25], [26], [27], [28], [30], yet is unable to provide valid and detailed information on the distribution and extent of dust coverage simultaneously. The research on Martian dust deposition thickness by the rover is mainly based on calibration data, and a few detection methods that utilize the ratio of red to blue bands of scientific data are also susceptible to shadow interference [20]. Therefore, these methods are incapable of accurately representing the spatial distribution of dust cover in small-scale Martian scientific observation scenarios.

To address the aforementioned limitations, this article proposes a novel in situ DCI (IDCI) approach based on MSCam observations. The main contributions and novelties are summarized as follows.

1) *Data-Driven Target Discrimination:* The proposed IDCI is a data-driven approach that pays particular attention to key spectral features indicative of the presence and degree of Martian dust coverage observed in the multispectral bands of MSCam. By exploiting these distinctive spectral features, it is feasible to optimize the contrast and distinguishability between different types of targets and levels of Martian dust cover in situ scenes. This data-driven approach ensures relatively accurate and meaningful mapping results, thereby enhancing the ability to identify regions of scientific interest.
2) *Efficient and Automatic Mapping:* The IDCI method's simple, yet effective band-math calculation, utilizing the original spectral bands, guarantees low computational complexity and high efficiency in comparison to techniques that employ intricate transformations, machine learning, or deep learning techniques. The method automatically determines thresholds from data distribution, identifying different types of targets and reflecting the degree of dust coverage based on index values. This automation eliminates the need for manual data selection or visual interpretation, significantly reducing the required human workload. These features make the IDCI suitable for handling large datasets and supporting real-time decision-making during rover missions, enabling rapid analysis and timely adjustments to optimize scientific exploration.
3) *High-Resolution Target Identification and Dust Coverage Degree Estimation:* In comparison to other payloads, MSCam is distinctive in its capacity for in situ exploration, offering integrated spectral and spatial information of Martian surface materials. The proposed IDCI method utilizes the high-resolution capabilities of MSCam's multispectral data to accurately identify areas on the Martian surface that are invalid, dust-covered, and dust-free. Furthermore, the IDCI offers continuous estimation of dust coverage in the areas with dust cover. This feature is of significant value in the selection of exploration targets with high scientific value and the exclusion of invalid areas during the exploration process.

TABLE I

KEY SPECIFICATIONS OF MSCam [36]

Items	Description
Bands and wavelengths	8 narrow bands: 480 nm, 525 nm, 650 nm, 700 nm, 800 nm, 900 nm, 950 nm, 1000 nm; 1 solar imaging band: 480 - 600nm.
Focal length	37.5 mm
Normal imaging distance	1.5 m ~ ∞
Effective pixel number	2K × 2K
Pixel size	5.5 μm × 5.5 μm
Field of View	24°
IFOV	0.146 mrad

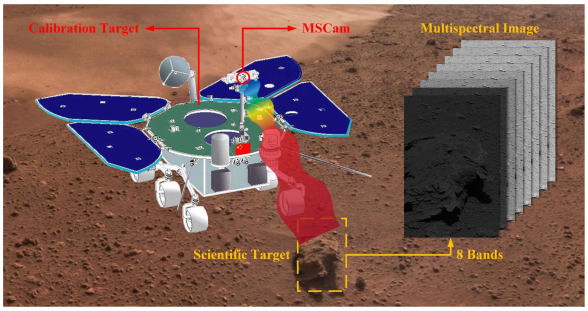


Fig. 1. Schematic of in situ multispectral exploration of the Martian surface by MSCam onboard the Zhurong rover, modified from [2].

The remainder of this article is organized as follows. Section II introduces the considered MSCam multispectral data. Section III provides a detailed description of the proposed IDCI approach. Section IV presents the experimental results and analysis. Finally, the conclusion is drawn in Section V.

II. DATA DESCRIPTION AND PREPROCESSING

MSCam, a scientific instrument aboard the Zhurong rover, is capable of simultaneously obtaining both spatial and spectral information, which makes it a significant tool for scientific exploration. The digital imaging system is equipped with a charge-coupled device (CCD) with a resolution of 2048 × 2048 pixels and is furnished with a filter wheel comprising eight narrowband filters and a broad solar band. Detailed information on the key parameters of the MSCam can be found in Table I. It is capable of capturing multichannel images across a range of wavelengths, from 480 to 1000 nm, in Mars exploration scenarios, as illustrated in Fig. 1. As shown in Fig. 2, the calibration target on MSCam mainly consists of three grayscale rings, each exhibiting a distinct reflectance value of approximately 20%, 40%, and 60%, respectively. A T-shaped gnomon is located at the center of the calibration panel, casting a shadow on the grayscale rings to distinguish between the solar direct irradiance and sky diffuse irradiance [36]. Before undertaking Martian surface detection on each new solar day, MSCam captures a series of calibration images utilizing all eight narrowband filters. This enables

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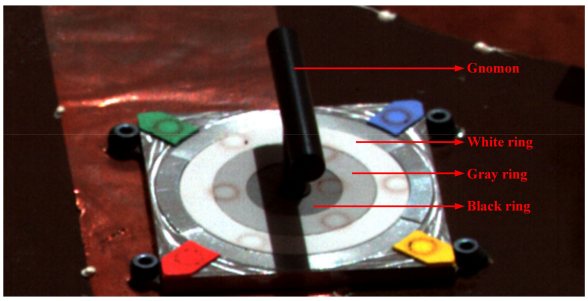


Fig. 2. Example image of the MSCam calibration target (Sol 43_00 data).

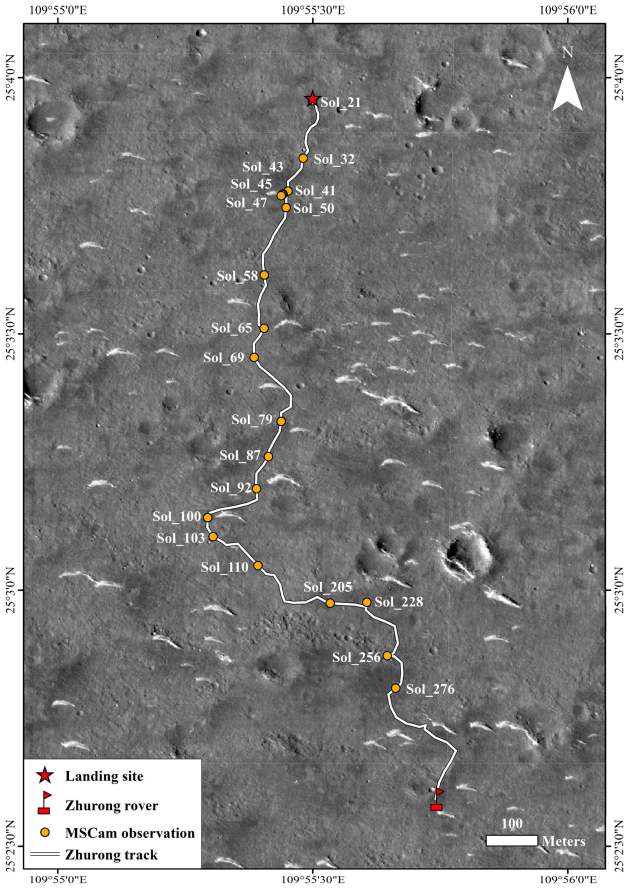


Fig. 3. Traverse of Zhurong rover on the Martian surface and the observation sites of MSCam. The background image is a mosaic of TW-1 HiRIC images (HX1_GRAS_HIRIC_DIM_0.7_0004_251515N1095850E_A).

the gathering of data pertaining to the proportion of direct and diffuse irradiance. Subsequently, MSCam proceeds with the acquisition of observational data for scientific exploration targets.

As of the 276th solar day (Sol) of Zhurong rover detection (February 21, 2022), MSCam had completed 36 scientific observations in 20 solar days. Fig. 3 shows the traverse path of the Zhurong rover and the locations of MSCam observation sites.

The objective of this study was to develop an index that is specifically designed for Martian rocky and dusty scenarios. Insufficient shadows on the calibration target can result in an inability to distinguish between direct and diffuse solar irradiance, which, in turn, affects the calculation of the

TABLE II
MSCam Data Filtering Results

Sol	No.	Type	Sol	No.	Type
21	05	♦	79	08	★
				09	★
32	05	♦	87	06	★
				07	★
41	06	●	92	06	●
	08	★		08	●
43	06	★	100	06	●
	07	★		08	★
45	06	■	103	06	★
	08	■		08	★
47	06	●	110	06	●
	07	●		08	★
50	08	■	205	06	♦
				07	♦
58	06	★	228	06	♦
	08	★		07	♦
65	06	●	256	05	★
	08	★			
69	06	★	276	04	●
	08	★		06	●

★ Valid; ♦ Insufficient shadow; ■ Incidence angle exceeded; ● Unrelated.

reflectance factor value and the estimation of dust cover. Considering the aforementioned factors, the following three types of data have been excluded from the final analysis: 1) observations with insufficient shadows on the three grayscale rings of the calibration target; 2) images captured at incident angles exceeding 45°, by taking into account the distribution of diffused light; and 3) scientific observation scenarios without typical rock or dust cover. This process enables the identification of valid observations and the detection of representative rock and dust deposition. Finally, observational data from 17 detection scenarios in 11 solar days were selected for analysis. Details of the data filtering results are provided in Table II.

In addition to multispectral scientific observations at each station, an imaging of the calibration target is conducted with a close observation time. Consequently, the calibration target images can be used in conjunction with the preflight data to transform the raw image data into radiance factor or I/F, where I represents the measured radiance and πF denotes the incident solar irradiance. The angle of solar incidence θ_i for each observation was recorded in a label file, and the data can be converted to reflectance factor (R^*, defined as I/F divided by the cosine of the incidence angle) according to the following equation (1) [37]:

$$R^* = \frac{I/F}{\cos\theta_i} \tag{1}$$

$$r_{\text{target}}(\lambda) = \frac{I_{\text{target}}(\lambda)}{k(\lambda)}. \tag{2}$$

According to [38], the preprocessing of 2B-level MSCam multispectral data involves several key steps. First, the digital number (DN) values are converted to radiance using the absolute radiometric calibration parameters provided in the image's label file. Next, pixels of uniform brightness are

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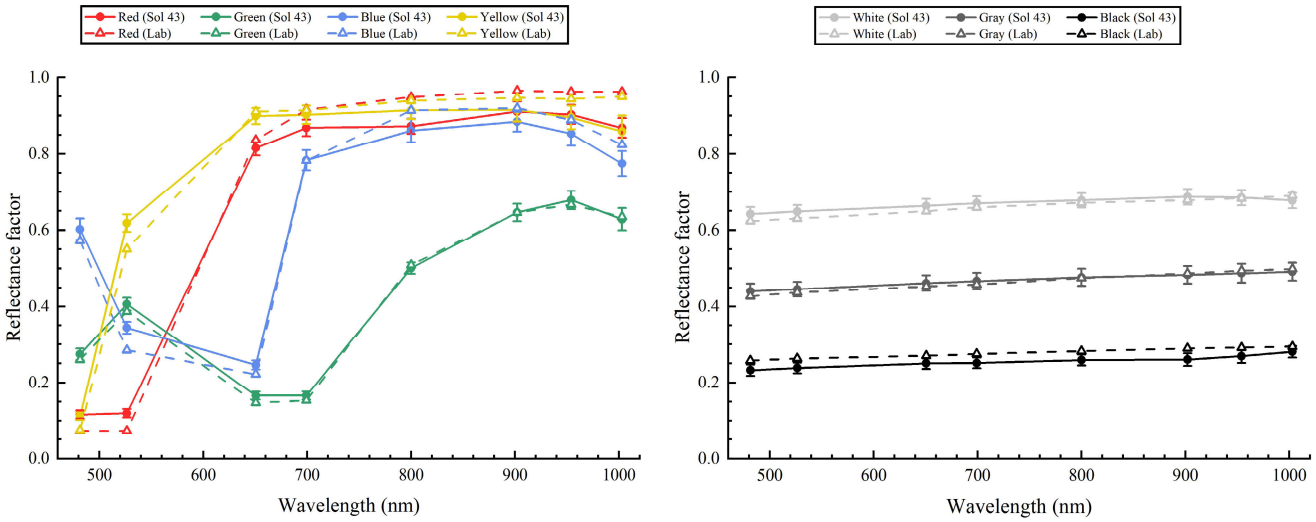


Fig. 4. In-flight reflectance calibration results of the MSCam calibration target (Sol 43_00 image in Fig. 2). (a) Directional-hemispherical reflectance factors of the four patches. (b) Weighted reflectance factors of the grayscale sunlit regions.

TABLE III
DESCRIPTION OF DIFFERENT REGION CATEGORIES IN THE DETECTION SCENE

Major category	Subcategory	Description
Dust-free	-	Mainly clean rock or soil
Dust-covered	Dust-covered L1	Slight dusty areas
	Dust-covered L2	Moderate dusty areas
	Dust-covered L3	Highly dusty areas
Invalid	-	Areas with shadows or data anomalies

selected as regions of interest (ROIs) in both the sunlight and shadow regions of the calibration target image and their average radiance values are determined. Then, these values are fit to laboratory-modeled reflectance factors using least-squares fitting, from which the reflectance factor (r_{target}) for scientific observations is derived according to (2).

Due to the limitation of downlink bandwidth, the complete image of each band (2048 × 2048) is partitioned into 64 discrete tiles. Finally, the subwindow images are merged based on their relative positions, and the single-band images are stacked into a multispectral image. This process ensures accurate radiometric calibration and prepares the data for subsequent analysis. Note that for observations after Sol 110, a dust correction model based on a two-layer radiative transfer Hapke model was employed to compensate for the accumulation of airfall dust on the calibration target during the mission [35]. The in-flight reflectance calibration result of MSCam calibration target (Sol 43_00 image in Fig. 2) is shown in Fig. 4, which confirms the effectiveness of radiometric calibration.

III. METHODOLOGY

A. Principle of the Proposed Index

To describe and analyze the characteristics of different dust coverage levels from a spectral perspective, a series of highly reliable ROIs were selected from several typical scientific scenarios observed by MSCam. For clarity, Table III delineates three major categories of ROIs: dust-free, dust-covered, and invalid. The dust-covered category is further divided into three subcategories: dust-covered L1, dust-covered L2, and dust-covered L3, with the corresponding degree of dust coverage increasing in sequence. The average spectra of each category were plotted in Fig. 5.

Through careful observation and comparison, we can find that the spectral characteristics influenced by the Martian dust cover are mainly characterized by two key aspects: the reflectance factor value and the band slope. Details are provided as follows.

1) *Reflectance Factor Value:* It is evident that the reflectance factor values of the invalid shadow areas are significantly lower than those observed in the other two major categories. In regions devoid of invalid shadow areas, the reflectance factor is higher in all bands when compared to regions covered with dust. This phenomenon indicates that the presence of Martian dust on rocks will result in a reduction of the reflectance factor value. A further comparison of the three spectral curves corresponding to ROIs of areas with different dust coverage reveals a consistent trend: as the dust coverage on the Martian surface increases, the reflectance factor values of the corresponding areas decrease.
2) *Band Slope:* The shape of the spectral curve can be described in terms of the band slope. By aligning the spectral curves of the five ROI categories at each band, it was found that in the first three bands, areas with different levels of dust coverage exhibited different band

$$k_{12} = \frac{B_2 - B_1}{\lambda_2 - \lambda_1} \tag{3}$$

$$k_{23} = \frac{B_3 - B_2}{\lambda_3 - \lambda_2} \tag{4}$$

$$S = \sum_{n=1}^{8} B_n \tag{5}$$

$$\text{IDCI} = (k_{12} + k_{23})S = \left(\frac{B_2 - B_1}{\lambda_2 - \lambda_1} + \frac{B_3 - B_2}{\lambda_3 - \lambda_2}\right) \sum_{n=1}^{8} B_n \tag{6}$$

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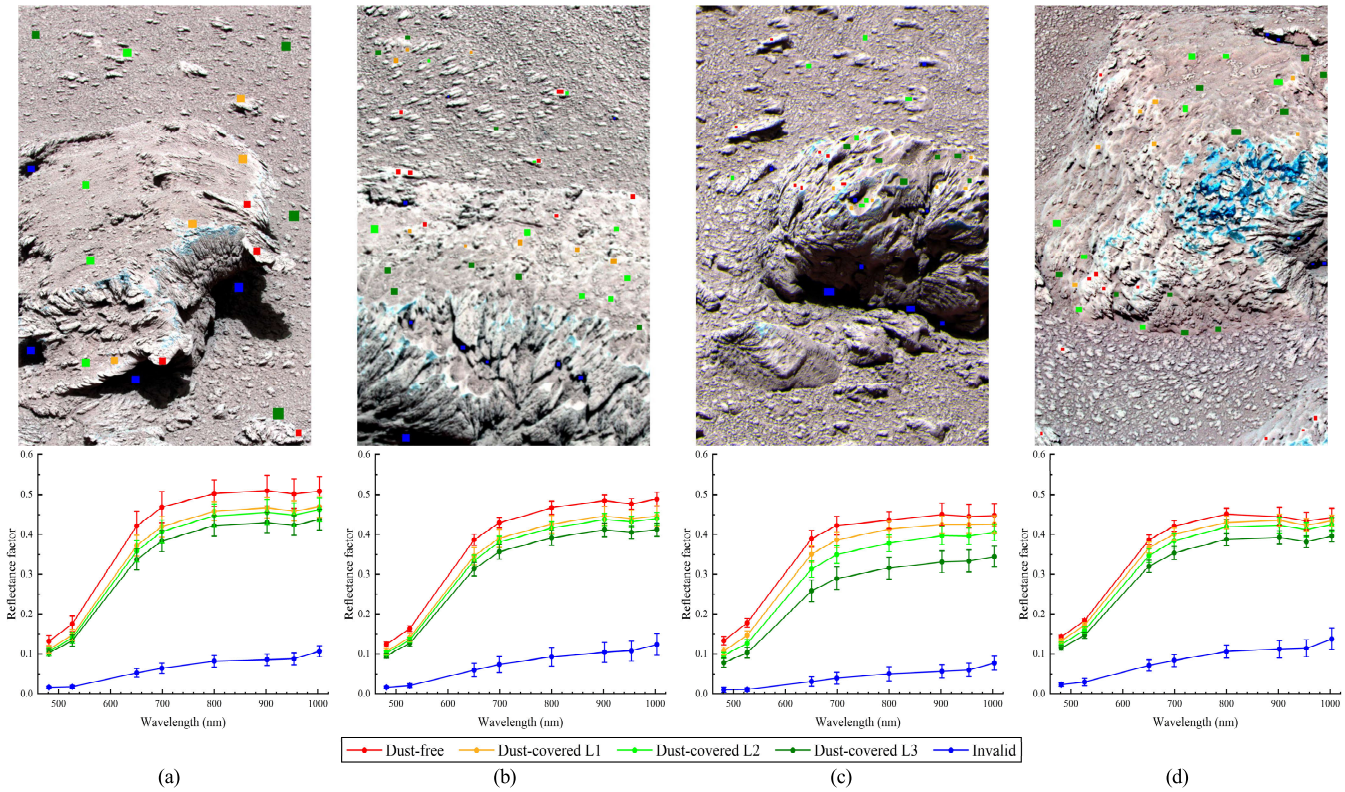


Fig. 5. ROI labeling and statistical results for five categories of samples from different scientific observations. The first row: true color images with ROIs; the second row: reflectance factor curves. (a) Sol 43_06 data, (b) Sol 79_08 data, (c) Sol 103_06 data, and (d) Sol 256_05 data.

slope characteristics. The spectral curve slope from 480 to 525 nm decreases with increasing dust coverage, while the slope of the invalid shadow area is the lowest. Furthermore, this trend is amplified in the range of 525–650 nm. Therefore, the band slope from 525 to 650 nm can also be employed for characterizing category differences.

In light of the aforementioned important observation and analysis of the law of Martian dust cover, a novel IDCI is proposed to identify scientifically valuable and invalid areas in the scene, while simultaneously reflecting the degree of dust coverage.

This index employs all eight bands of MSCam and is constructed using the band slope and R^* values. Let the MSCam reflectance factors input be $\{B_1, B_2, B_3, B_4, B_5, B_6, B_7, B_8\}$, including visible to near-infrared bands. To distinguish between dust-free and dust-covered regions, two slopes were calculated, and the sum of them was used as a coefficient. The slope k_{12} of the 480–525 nm band interval is calculated according to formula (3), and the slope k_{23} of the 525–650 nm band interval is calculated according to formula (4). Among them, B_1, B_2, and B_3 are the reflectance factor values on the blue, green, and red bands, respectively, and λ_1, λ_2, and λ_3 are the central wavelength values (measured in nanometer) corresponding to the blue, green, and red bands. To highlight the spectral distinction between invalid regions and valid ones, the sum S of the reflectance values for each band captured by MSCam was calculated and obtained according to (5). Finally, the IDCI is generated in accordance with the specifications outlined in (6).

B. Target Identification and Dust Coverage Estimation

Once the IDCI results have been obtained, the threshold for classifying the different categories is determined based on the data distribution, thereby achieving the identification of high-value scientific targets. Initially, the mean value μ_{IDCI} and standard deviation value σ_{IDCI} of the IDCI map are statistically acquired. Subsequently, the lower limit T_1 and upper limit T_2 that define the dust-covered area are calculated according to the following equations:

$$T_1 = \mu_{\text{IDCI}} - \sigma_{\text{IDCI}} \tag{7}$$

$$T_2 = \mu_{\text{IDCI}} + \sigma_{\text{IDCI}}. \tag{8}$$

The images of the Martian scene can be classified into three categories of target regions based on the thresholds T_1, T_2, and IDCI. For a given pixel p in the image, its corresponding IDCI value IDCI_p determines its category

$$p \in \begin{cases} \text{Invalid}, & \text{IDCI}_p < T_1 \\ \text{Dust-covered}, & T_1 \le \text{IDCI}_p \le T_2 \\ \text{Dust-free}, & \text{IDCI}_p > T_2. \end{cases} \tag{9}$$

For areas that are identified as dust-covered, the IDCI can be employed to provide an estimation of the dust coverage degree. A larger IDCI value indicates a lower degree of dust coverage, and vice versa. The final in situ dust cover mapping results include two output images.

1) *A Scene Target Classification Map:* It is a thematic map with three-category identified targets, representing dust-free, dust-covered, and invalid areas, respectively.

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2) *A Dust Coverage Estimation Map:* Different levels of dust coverage are illustrated, where the yellow end indicates low dust coverage, while the blue end represents high dust coverage.

The whole process of the proposed method is summarized in Algorithm 1.

Algorithm 1 In Situ Dust Cover Index Calculation and Thematic Maps Generation

Input: 2B-level Multispectral image data MSI
Output: Scene target classification map I_c and dust coverage estimation map I_e

Obtain reflectance factor $\{B_1, B_2, ..., B_8\}$ by applying radiometric calibration to MSI
Calculate band slopes $k_{12} = \frac{B_2 - B_1}{\lambda_2 - \lambda_1}$ and $k_{23} = \frac{B_3 - B_2}{\lambda_3 - \lambda_2}$
Calculate sum of reflectance factor values $S = sum(B_1, B_2, ..., B_8)$
Output the in situ dust cover index $IDCI = (k_{12} + k_{23}) * S$
Obtain the mean μ_{IDCI} and standard deviation σ_{IDCI} of $IDCI$
Calculate thresholds $T_1 = \mu_{IDCI} - \sigma_{IDCI}$ and $T_2 = \mu_{IDCI} + \sigma_{IDCI}$
foreach *pixel p in the image MSI* **do**
    if $IDCI_p < T_1$ **then**
        $p \in$ Invalid;
    else
        if $IDCI_p > T_2$ **then**
            $p \in$ Dust-free;
        else
            $p \in$ Dust-covered;
Generate scene target classification map I_c = Classification($IDCI$);
Generate dust coverage estimation map I_e = Estimation($IDCI$);
final;
return I_c, I_e;

IV. Results and Analysis

A. Experimental Setup

To evaluate the effectiveness of the proposed IDCI, it was compared with four other popular methods, including: 1) red/blue ratio (R/B); 2) DCS; 3) PCA; and 4) MNF. For a detailed quantitative evaluation, two numeric indicators were calculated. First, the Calinski–Harabasz index, known as the variance ratio criterion (VRC) [39], was calculated to reflect the multiclass separability of different methods. This indicator is mainly composed of within-group sum of squares (WGSSs) and between-group sum of squares (BGSSs) [see (10) and (11)], with the overall calculation process similar to that of variance. Essentially, it is the ratio of intercluster distance to intracluster distance. Let N be the total number of samples, and K be the number of clusters. N_i denotes the number of samples in cluster i, C_i represents all samples in cluster i, x is a sample belonging to C_i, c_i is the center of cluster i, and c is the global cluster center. The calculation formula for VRC is provided in (12).

A higher VRC value indicates that the BGSS is considerably much larger than the WGSS. In this case, the intercluster distances are significantly greater than the intracluster distances, indicating that the clusters are effectively separated and that the clustering result is optimal. Conversely, a lower VRC indicates a relatively poor clustering result. Note that the range of the VRC is $(0, +\infty)$

$$\mathrm{WGSS} = \sum_{i=1}^{K} \sum_{x \in C_i} (x - c_i)^2 \tag{10}$$

$$\mathrm{BGSS} = \sum_{i=1}^{K} N_i (c_i - c)^2 \tag{11}$$

$$\mathrm{VRC} = \frac{\mathrm{BGSS}}{K-1} \Big/ \frac{\mathrm{WGSS}}{N-K}. \tag{12}$$

Second, the computation time (TC) was employed to assess the computing complexity and efficiency of reference methods. The specific TC values were obtained by recording the average running time of the code in MATLAB software. The algorithms described in this article were implemented on a hardware system comprising an Intel Core i5-12500H CPU, 16 GB of RAM, and an NVIDIA RTX 2050 GPU with 4 GB of graphics memory. The software implementation was conducted on a Windows 11 operating system using MATLAB and PyCharm software.

B. In Situ Dust Cover Mapping Results

As shown in Fig. 6, the first row represents the original true color composite images. It is evident that there are several rocks present in the scene, exhibiting varying degrees of dust coverage, and the background is identified as Martian soil. Images in the second row are scene target classification maps, where the invalid areas (e.g., shadows), dust-covered areas, and dust-free areas are black, brown, and red, respectively. Images in the third row represent a dust coverage estimation map, where pixels in different colors indicate the varying degrees of dust coverage.

The high and low values of IDCI represent dust-free and dusty surfaces, respectively. The series of intermediate IDCI values appearing on the map are spatially continuous, indicating that they may represent surfaces obscured by varying degrees of dust. It is possible that the presence of dust may reduce the reflectance of the material, resulting in this continuous IDCI value.

C. Qualitative Analysis in Comparison With Reference Methods

Qualitative analysis was conducted by visually comparing the obtained scene target classification maps. Fig. 7 presents a comparative analysis of the results obtained by five algorithms on the Sol 87_06 data. The rectangle boxes in the images are highlighted for comparison purposes. It is important to note that in the dust coverage estimation maps of DCS, PCA,

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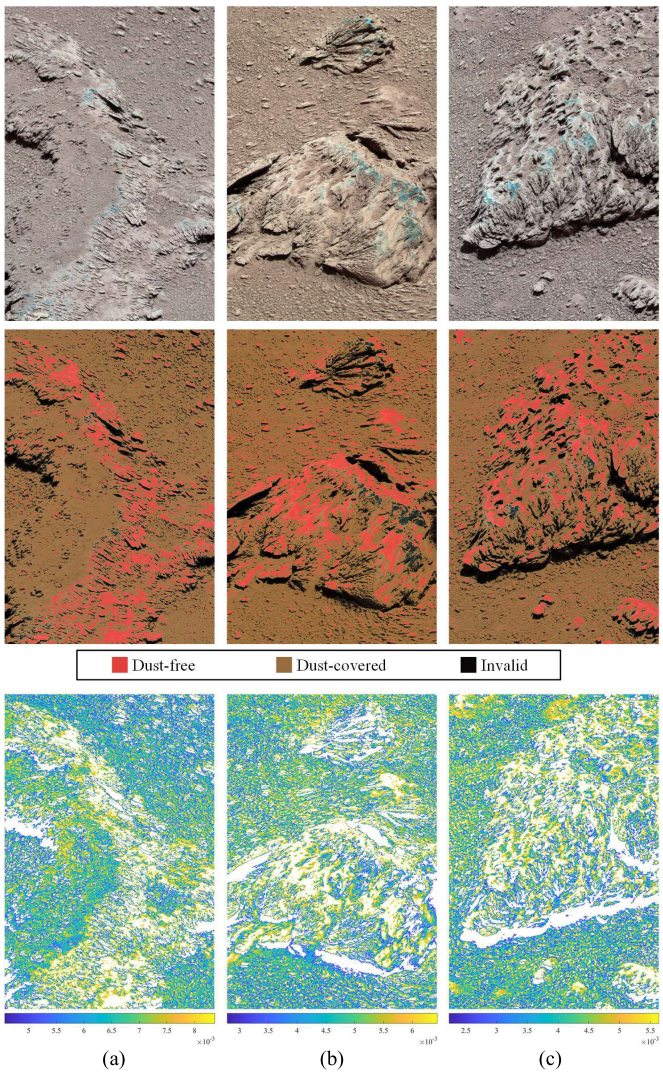


Fig. 6. In situ dust cover mapping results of different observations. First row: true color images of the observed scenarios; second row: scene target classification maps, using true color images of the observed scene as the background image, with 40% transparency of the labeled colors (red, brown, and black) for different categories; third row: dust coverage estimation maps. (a) Sol 43_07 data, (b) Sol 79_09 data, and (c) Sol 87_06 data.

and IDCI, larger values indicate lower dust coverage and are closer to the yellow end of the color bar, while smaller values indicate a higher coverage range, and are closer to the blue end of the color bar. In contrast, the R/B and MNF maps illustrate an inverse relationship between color and value. To ensure consistency in color interpretation, the color bars were inverted, with the yellow end representing smaller values and the blue representing larger values.

The R/B results for Sol 87_06 exhibited a significant increase in noise compared to other methods. Additionally, a large number of pixels in shadow and rock outcrop regions lacking dust cover were incorrectly identified and classified as dust-covered areas [see Fig. 7(a)]. The DCS results are in close alignment with the IDCI, yet a notable limitation is that the DCS results may underestimate the degree of dust coverage [see Fig. 7(b)]. With regard to the PCA method, it should be noted that some of the identified dust-free areas

TABLE IV
COMPUTATION TIME RESULTS OF DIFFERENT METHODS

Methods	Computation Time (ms)
R/B	**44.8**±15.1
DCS	98.6±23.0
PCA	166.8±35.1
MNF	108.2±20.0
IDCI	53.2±**13.3**

TABLE V
VRC RESULTS OF DIFFERENT METHODS

Methods	VRC ($\times 10^5$)
R/B	4.42±1.28
DCS	6.02±1.59
PCA	5.99±1.68
MNF	5.90±1.51
IDCI	**6.30**±**1.07**

are spatially incomplete and discontinuous [see Fig. 7(c)]. The main issue with the MNF results is that the regions covered by dust are cut very finely, resulting in a lack of continuity in estimating the degree of dust coverage [see Fig. 7(d)]. Compared with the results of other methods, IDCI thematic maps demonstrate an improved discriminative effect on the three classes of target regions. By utilizing the proposed index, a spatially and numerically continuous coverage estimation was readily achieved for the dust-covered area [see Fig. 7(e)].

In the Sol 79_09 image scenario, two rocks are clearly discernible. From the obtained maps, one can observe that both the R/B and MNF methods continue to exhibit a significant limitation of high-level noises, making it difficult to clearly identify different categories of target regions within the scene and accurately represent the dust coverage. The estimation of dust coverage by DCS and PCA is found to be relatively lower than the actual situation. However, the proposed IDCI method demonstrates superior performance in identifying dust-free and invalid areas and offers an accurate estimation of dust coverage [see Fig. 8(e)].

In the Sol 43_07 image, a relatively large flat rock was present in the scene. As shown in Fig. 9, the results obtained by the reference methods exhibited a more pronounced noise phenomenon and an underestimation of dust coverage compared to the proposed IDCI method.

The visual evaluation of the three MSCam observation scene cases revealed that the classification maps and dust coverage estimation results obtained by IDCI achieved superior performance compared to other reference methods.

D. Quantitative Analysis in Comparison With Reference Methods

To provide a further comprehensive evaluation of the performance of considered algorithms, quantitative statistics were conducted on MSCam data acquired from 17 scenarios along the routing path of the Zhurong rover. Numeric results are presented in Tables IV and V, which show the computation

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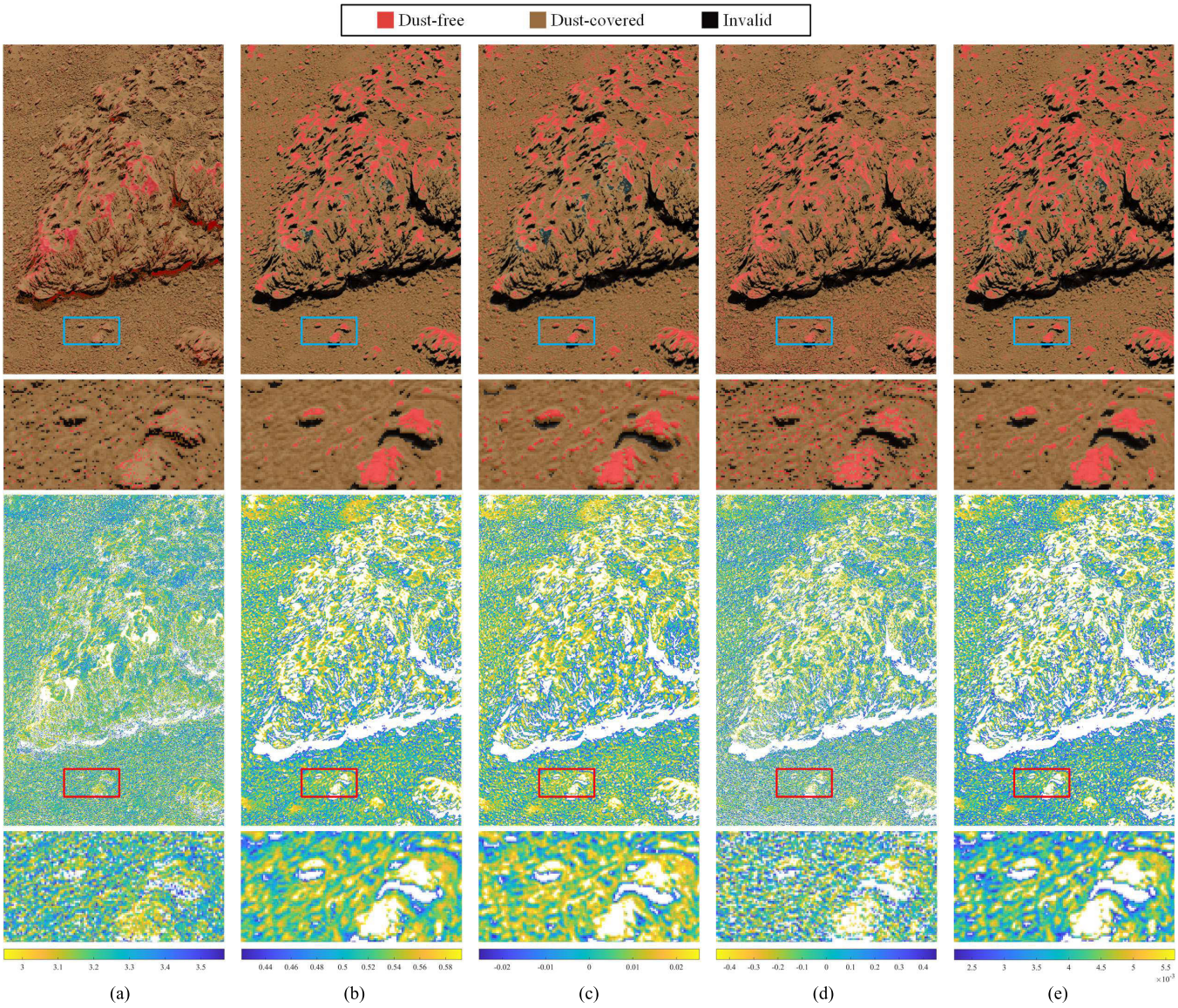


Fig. 7. In situ dust cover mapping results obtained by different methods on Sol 87_06 multispectral data. First row: scene target classification maps; second row: enlarged subsets of the blue rectangle that are highlighted in the scene target classification maps; third row: dust coverage estimation maps; fourth row: enlarged subsets of the red rectangle that are highlighted in the dust coverage estimation maps. (a) R/B, (b) DCS, (c) PCA, (d) MNF, and (e) IDCI.

time and VRC of different methods, respectively. In the tables, the best values of the evaluation criteria are highlighted in bold and the second-best values are underlined. Figs. 10 and 11 show the bar chart of computation time and box plot of VRC, respectively.

As shown in Table IV, the performance of each algorithm can be comprehensively evaluated based on the computation time of considered five different methods. Among all methods, the IDCI method ranks second in terms of average time consumption (i.e., 53.2 ms). In contrast, PCA is the slowest method, with an average computation time of 166.8 ms, which is approximately 3.1 times that of IDCI. The runtime of MNF (i.e., 108.2 ms) and DCS (i.e., 98.6 ms) is approximately 2 and 1.9 times that of IDCI, respectively. The R/B ratio method is the most efficient one, with an average computation time of 44.8 ms, which is slightly more rapid than the average time for IDCI. However, the R/B exhibits a higher standard deviation of 15.1 ms in comparison to IDCI (i.e., 13.3 ms), which indicates a more considerable variability in performance. In the case where the R/B method only relies on the red and blue bands in the original dataset, the IDCI is observed to achieve comparable computational time to that of the R/B method, highlighting its notable computational efficiency. The efficiency of the IDCI method can be attributed to its use of simple arithmetic operations on the eight original spectral bands of MSCam, avoiding the complex band transformations required for DCS, PCA, and MNF. Meanwhile, the simple, yet efficient mathematical operations employed by the IDCI also ensure that its computational time is less influenced by the size of the dataset, thus enhancing the stability of its performance in experimental settings. In conclusion, the IDCI method represents an optimal balance between speed and stability. Its average computation time, which is slightly higher than that of the R/B ratio, is demonstrably lower than

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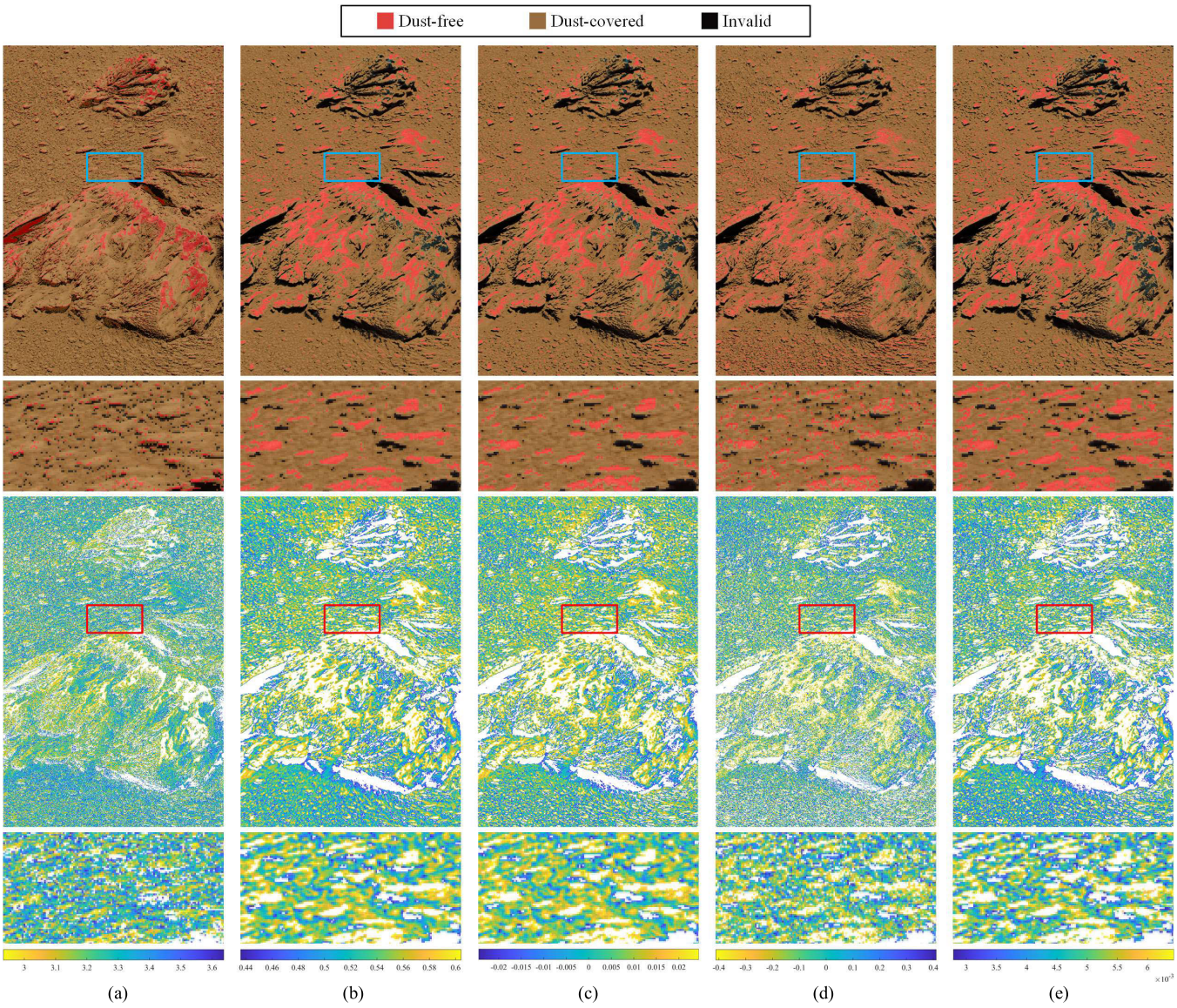


Fig. 8. In situ dust cover mapping results obtained by different methods on Sol 79_09 multispectral data. First row: scene target classification maps; second row: enlarged subsets of the blue rectangle that are highlighted in the scene target classification maps; third row: dust coverage estimation maps; fourth row: enlarged subsets of the red rectangle that are highlighted in the dust coverage estimation maps. (a) R/B, (b) DCS, (c) PCA, (d) MNF, and (e) IDCI.

that of DCS, PCA, and MNF. Furthermore, IDCI exhibits the lowest standard deviation values, indicating a more stable performance when dealing with varying complex scenarios. This makes the IDCI method superior for efficient and reliable data processing, outperforming DCS, PCA, and MNF in terms of both efficiency and stability.

In addition to the computation time, the VRC is also employed as a further means of comparison with respect to the performance of the aforementioned methods. The VRC results are presented in Table V.

As we can see, the IDCI method demonstrates superior performance, as indicated by a VRC value of 6.30×10^5, which is higher than that of DCS (i.e., 6.02×10^5) and PCA (i.e., 5.99×10^5), MNF (i.e., 5.90×10^5), and R/B (i.e., 4.42×10^5) on three target classes. Furthermore, the minimal standard deviation value of IDCI's VRC (i.e., 1.07×10^5) highlights the method's stability and reliability in distinguishing the three classes. In comparison, other methods, such as R/B, DCS, PCA, and MNF, exhibit higher standard deviation values, indicating greater variability in the performance of target separation. The superior performance of the IDCI method in terms of the quantitative separation index can be attributed to two main factors. First, the IDCI method amplifies the spectral differences between regions with varying dust coverage by using the sum of slopes between two band intervals as a multiplicative factor. This enhances the separation between these different categories. Second, the IDCI considers that the reflectance in all eight bands of the MSCam is typically lower in shadowed areas than in nonshadowed areas. Therefore, by summing the reflectance across all bands, the IDCI more effectively highlights the differences in reflectance between nonshadowed and shadowed areas, resulting in better differentiation between them.

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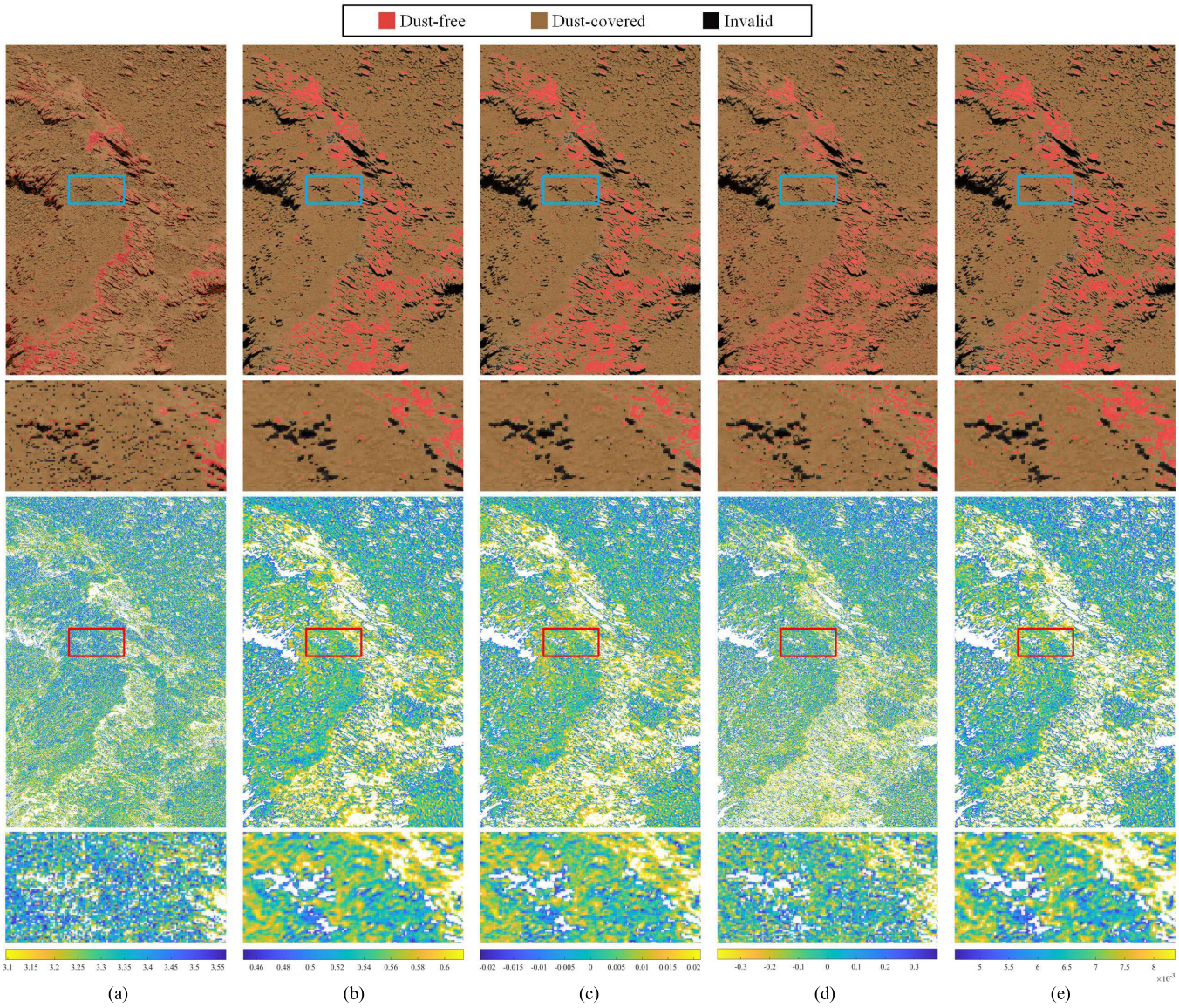


Fig. 9. In situ dust cover mapping results obtained by different methods on Sol 43_07 multispectral data. First row: scene target classification maps; second row: enlarged subsets of the blue rectangle that are highlighted in the scene target classification maps; third row: dust coverage estimation maps; fourth row: enlarged subsets of the red rectangle that are highlighted in the dust coverage estimation maps. (a) R/B, (b) DCS, (c) PCA, (d) MNF, and (e) IDCI.

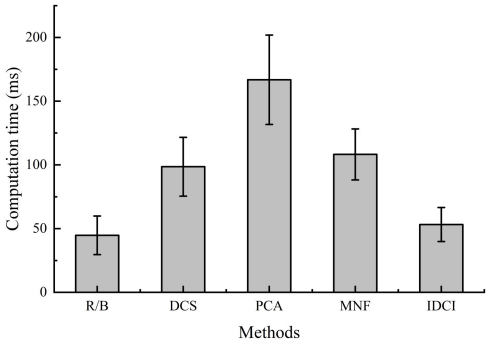


Fig. 10. Comparison of the computation time of different methods.

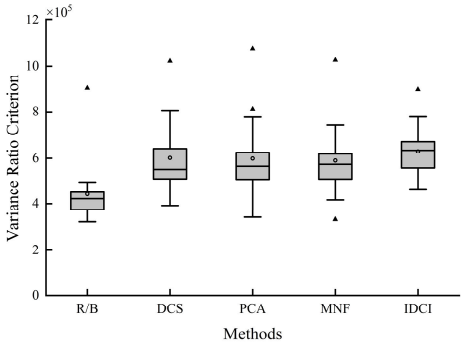


Fig. 11. Comparison of the VRC results obtained by different methods.

Furthermore, 3-D scatter plots of the classification results obtained by different methods were compared, as shown in Fig. 12, where a feature space R^3 (consisting of an index band, an average band of all multispectral bands, and the third band of the multispectral image as three axes). Given the considerable number of samples, the scatter points were randomly selected from 1% of the overall pixels. One can see from Fig. 12 that the degree of cluster concentration of the scatter

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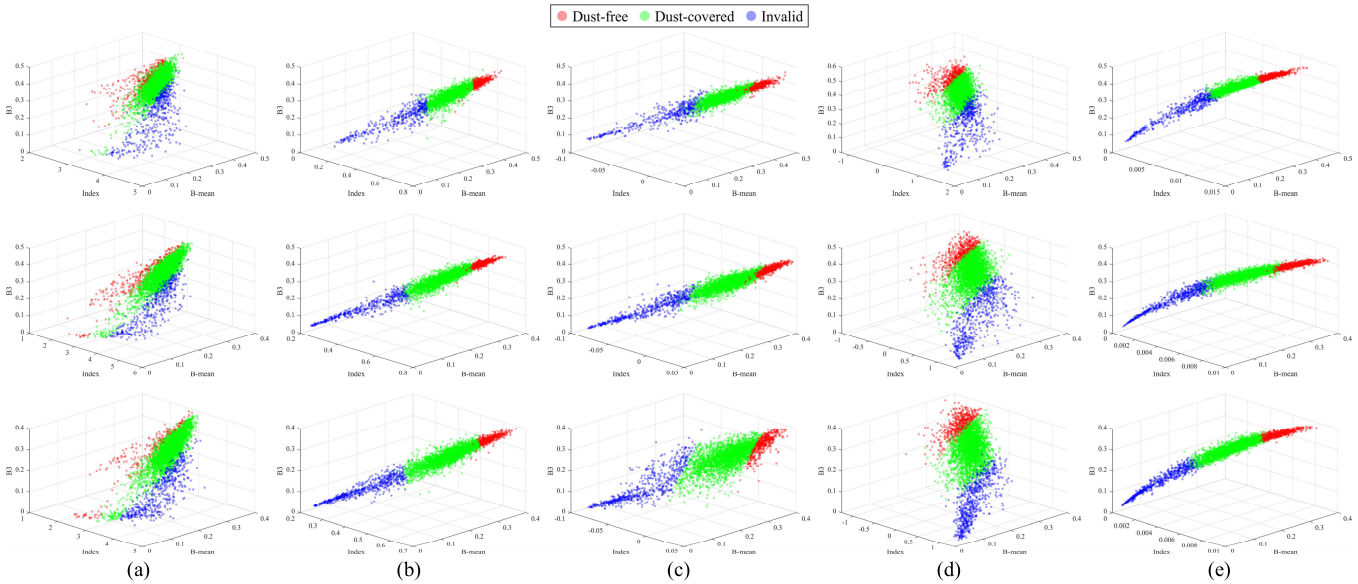


Fig. 12. Scatterplot of scene target classification results obtained by different methods. First row: Sol 43_07 data; second row: Sol 79_09 data; third row: Sol 87_06 data. (a) R/B, (b) DCS, (c) PCA, (d) MNF, and (e) IDCI.

plot of the proposed IDCI method significantly outperforms the R/B and MNF methods. It exhibits more concentrated clusters than the compared PCA and DCS methods, indicating its effectiveness for the feature representation of three target classes.

In summary, the IDCI method not only demonstrates efficient computation time but also exhibits the highest and most stable VRC value, thereby showcasing its effectiveness and reliability in distinguishing areas with minimal dust coverage, significant dust coverage, and invalid shadows on Mars. The IDCI method outperforms other reference methods by providing a superior balance between efficiency and accuracy, making it an ideal choice for studying the in situ dust cover issues on the Martian surface.

V. Conclusion

In this article, a novel IDCI was proposed, which offers a novel perspective for Martian surface dust analysis. The approach is proposed based on the physical spectral attributes exhibited in MSCam observations, rather than data transformation or stretch. It realizes an automated and high-resolution identification of invalid, dust-covered, and dust-free areas, improving data-processing efficiency and reliability. Additionally, the IDCI method provides continuous dust coverage estimation, which is crucial for understanding the varying degrees of dust deposition on the Martian surface before the material identification. The IDCI method demonstrates an optimal balance between processing efficiency and accuracy, making it suitable for real-time analysis and operational decision-making during rover in-flight missions.

Experiment results obtained on the real TW-1 MSCam dataset confirm the effectiveness of the IDCI method in identifying scientific targets and estimating dust coverage. A comprehensive qualitative and quantitative evaluation demonstrates that the IDCI algorithm outperforms many existing methods in terms of both computation time and the VRC for category separation. Future research will aim to extend the applicability of the IDCI method by applying it to multispectral data from other payloads (e.g., Mastcam and Mastcam-Z). In addition, considering the interference of Martian dust deposition on the reliability of material detection, further research will be conducted on the dust-free area identified by the IDCI map. This will entail an analysis of the material composition along the Zhurong rover's routing path, by incorporating the observation from other in situ instruments, such as the Mars Surface Composition Detector (MarSCoDe) [40], [41]. The detection of dust-covered areas will contribute to a more comprehensive understanding of dust coverage and deposition on Mars. Furthermore, the identification of dust-covered regions that may potentially impact the quality of observational data will assist in the planning of future in situ exploration missions.

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